

Saba Yazdani

Software Developer | Java, Python, JavaScript | REST APIs, PostgreSQL, CI/CD

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SUMMARY

Software developer (B.Sc. Computer Science, GPA 3.84) who builds and maintains Gradify, a production platform used across York University courses: **Java/Spring Boot RESTful APIs**, **Python** services, JavaScript/TypeScript front-ends, and **PostgreSQL**. Works with **Git**-based **CI/CD**, unit testing, debugging, code reviews, and documentation.

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript (ES6+), TypeScript, SQL, C, C#
- **Back-End & APIs:** RESTful APIs, Spring Boot, FastAPI, Node.js, .NET, OAuth2, SSE
- **Databases:** PostgreSQL, MySQL, SQLite (relational); Neo4j (NoSQL)
- **Cloud & DevOps:** Git, GitHub Actions (CI/CD), Docker, Google Cloud (GCP: Cloud Run, Functions), Azure, AWS EC2, Linux
- **Practices:** Agile/Scrum, Software Development Life Cycle (SDLC), Unit & Integration Testing (JUnit 5, pytest), Code Reviews, Debugging, Technical Documentation
- **Front-End:** React, Next.js, TypeScript, HTML/CSS, JavaFX
- **AI & Data:** LLM Applications, Structured Outputs, RAG, Vector Search (FAISS), PyTorch, scikit-learn, Pandas, NumPy

WORK & RESEARCH EXPERIENCE

Research Assistant

York University

Apr 2024 – Present

Toronto, ON

• Programming Education Lab

Sep 2024 – Present

- Built and maintain **Gradify**, the lab's grading platform (**Java**, Spring Boot, React), in production use by instructors and students across senior programming courses.
- Wrote **unit and integration tests** covering grading logic and correctness; debugged and resolved issues in the deployed system during live courses.
- Managed releases through **Git**-based workflows; co-authored **three peer-reviewed conference publications** on the platform.

• Data Mining Lab

Jan 2025 – Dec 2025

- Developed a **Python** geospatial question-answering service using FAISS vector search and a lightweight LLM to answer natural language queries over structured spatial datasets.
- Built data preprocessing and feature extraction pipelines; evaluated outputs for accuracy, consistency, and reliability.
- Wrote the project's **technical documentation** and co-authored the resulting research paper.

• Theory Computing Lab

Apr 2024 – Present

- Built succinct graph data structures enabling compact storage and efficient queries on large-scale datasets.
- Implemented and evaluated algorithms, analyzing performance, correctness, and memory trade-offs.
- Co-authored the peer-reviewed paper “Geometric Burning Under L_1 and L_∞ Metrics, and Beyond,” accepted at WAOA 2026.

Teaching Assistant

Jan 2025 – Dec 2025

Led lab sessions, ran **code reviews** of student work, and provided debugging support and grading across EECS2021 (Computer Organization), EECS2030 (Object-Oriented Programming), and EECS3000 (Professional Practice).

PROJECTS

Gradify – AI-Assisted Grading Platform

Java, Spring Boot, React, PostgreSQL, REST APIs, JUnit 5, Checkstyle, Claude, Gemini

- Built and deployed the **full stack** for live York University courses: React instructor, student, and admin workflows over **Spring Boot RESTful APIs**, **PostgreSQL** persistence, and role-based access control.
- Built a **Java** autograding pipeline with in-memory compilation (JavaCompiler API), **JUnit 5** execution via the Platform Launcher, and Checkstyle static analysis with weighted test and style scoring.
- Developed asynchronous batch processing with Spring **@Async** and custom thread pools: concurrent multi-student grading, per-batch locking, retries, progress tracking, and email delivery.
- Integrated Claude and Gemini REST APIs for multimodal grading of handwritten and PDF submissions, parsing structured JSON into rubric scores, instructor overrides, and CSV/PDF reports.

Realest – AI Voice Agent for Rental Verification

AI Tinkerers Agents, Everywhere Hackathon 2026

GitHub • *Python, FastAPI, Pydantic, asyncio, Next.js, React, ElevenLabs, Twilio, OpenAI, Gemini, GitHub Actions*

- Built, in a team of four, a voice-agent platform that captures renter preferences, builds a live shortlist, and calls listing agents to verify availability, fees, parking costs, and pet policies.
- Developed the **Python/FastAPI** backend for concurrent outbound calls using **asyncio**, webhook-based event handling, session-state management, per-call timeouts, and fallback logic.
- Owned the email fallback for unanswered or out-of-hours calls: SMTP delivery with recipient validation and a duplicate-send guard so no message goes out twice.
- Implemented an LLM transcript-processing pipeline using structured outputs to convert calls into validated `CallOutcome` data and re-rank listings.
- Debugged and fixed integration regressions across teammates' branches, kept the pytest suite passing, and automated health checks in GitHub Actions (**CI/CD**).

Maridian – Multi-Agent Decision Layer for the Factory Floor

TMLS Agentic Hackathon 2026

GitHub • *Next.js, React, FastAPI, Python, SQLite, Server-Sent Events, Pydantic, Vercel, Railway*

- Built a platform that ingests bakery production-line defect events and recommends recovery actions to floor operators.
- Developed the Python/FastAPI backend and its **REST API**, streaming live workflow updates to a React/Next.js dashboard over Server-Sent Events; deployed both tiers to managed **cloud** hosting.
- Engineered an LLM multi-agent workflow for incident triage, action planning, and scoring, with human-in-the-loop approval controls and decision logging so no customer-facing action ships without operator sign-off.

Systems Programming Projects

- Implemented FCFS, SJF, Priority, and Round-Robin CPU schedulers in C, and a virtual memory simulator with TLB, page table, and backing-store paging; evaluated turnaround and response times, TLB hit rates, and page-fault behavior.

EDUCATION

B.Sc. Computer Science, Specialized Honours, York University

2021-2025

GPA 3.84 / 4.0 • First Class with Distinction

Core Courses: Software Design (A+), Database Management Systems (A+), Design and Analysis of Algorithms (A+), Big Data Systems (A+), Artificial Intelligence (A+)